

Bitcoin Volatility Prediction

Forecasting 2-day realized volatility, not price direction

79,000+ Hourly BTC/USDT Records (2017-2026)

How Bitcoin's Price Swings

Big rallies, sharp crashes, and everything in between

Bitcoin's price is highly non-stationary: since 2017 it has drawn down more than 80% twice, then rallied to new all-time highs both times. These swings - not day-to-day direction - are the volatility this project forecasts.



WHAT THIS MEANS

These price swings are volatility - the dispersion of returns, not their direction. This project forecasts the magnitude of that dispersion 2 days ahead.

What Is Volatility, and Why Predict It?

Volatility measures the dispersion of price movements over time, independent of direction. Low volatility means small, tight moves; high volatility means large, rapid swings either way.

AN ANALOGY

Predicting price is like guessing exactly where a boat will be tomorrow - nearly impossible. Predicting volatility is like guessing how rough the sea will be. That's a more tractable question, and usually the more useful one for risk management.

Why Predicting It Matters

Who Uses It	Why It Matters
Options traders	Required input for Black-Scholes σ
Risk managers	Scales Value-at-Risk estimates
Traders	Position sizing under changing risk
Exchanges	Sets margin/collateral requirements

HOW IT'S CALCULATED

$$\sigma_{2d} = \sqrt{\frac{1}{N-1} \sum_{i=t-N+1}^t (r_i - \bar{r})^2}$$

r = hourly return, $\bar{r}$ = its average, N = 48 hours (2 days)

INTUITION

The standard deviation of the last 48 hourly returns. A wider spread around the mean means a more turbulent 2 days; a tighter spread means a calmer one.

Problem Statement

From a vague question to a testable target

THE AIM

Predict Bitcoin's 2-day realized volatility, 48 hours ahead of the features used to predict it - not the next hour's value. This gives traders and risk desks a continuously-updating volatility estimate, usable directly in position-sizing and margin decisions.

Target Definition

Target	volatility_2d, 48h ahead of the features
Horizon	48 hours (2 days) - never the next hour
Features	Lagged price/volume history
Success bar	Beat GARCH and naive persistence

A NUMBER, NOT A CATEGORY

Downstream uses - option pricing, Value-at-Risk, margin - need a continuous estimate, not a high/low label. That rules out classification: this is a regression problem.

Data Overview & Feature Dictionary

DATASET: 79,000+ hourly BTC/USDT records

FEATURES: Price, volume, engineered volatility measures, and calendar

TARGET: Volatility over the next 2 days

Feature	Description	Type
RV_day / RV_week / RV_month	Realized volatility over the past day/week/month (HAR horizons)	Numeric
ATR_X	Average True Range, as a % of price	Numeric
BB_width	Bollinger Band width, as a % of the moving average	Numeric
dow_sin / dow_cos	Cyclical day-of-week encoding (confirmed significant, see Feature Selection)	Numeric
hourly_return	Hour-over-hour % price change	Numeric
volatility_2d (target)	2-day realized volatility, 48h ahead of the features	Target

Dataset & Data Cleaning

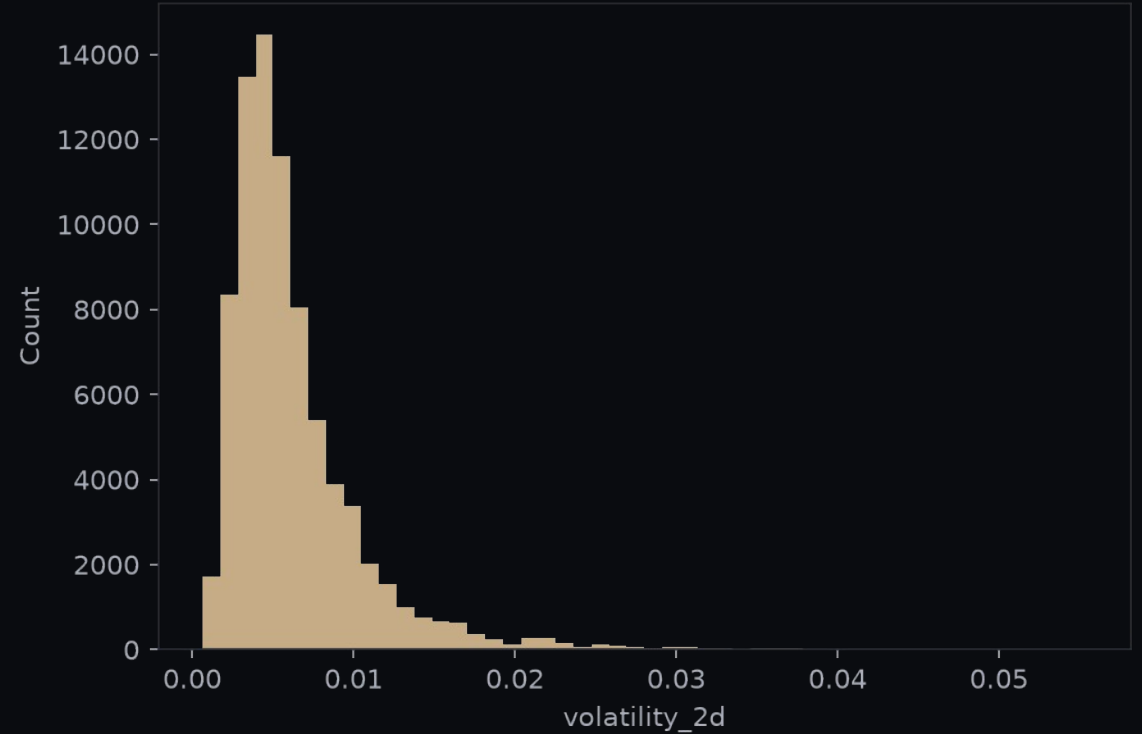
Data Cleaning Steps

1 Fixed 29 timestamp gaps in the combined 2017-2026 hourly series

2 Merged historical and live-collected data into one continuous file

3 Standardized hourly timestamp frequency, zero duplicates

4 Dropped ~30 days of rolling-window warm-up rows



Regime Mapping

Level	Regime	Volatility Range
0	Low	≤ 0.0041
1	Moderate	0.0041 - 0.0065
2	High	> 0.0065

Feature Engineering: HAR Horizons & ATR_X

The core volatility-based features

HAR Horizons (RV_day / RV_week / RV_month)

WHAT IT IS & WHAT IT MEANS

RV_week = mean(volatility_2d, past 168h). Same idea at 24h and 720h windows (Corsi's HAR model, 2009). Volatility has memory at multiple timescales - a rough day in a calm month differs from one in an already-turbulent month, and a single window can't tell those apart.

WHY WE NEED IT

Confirmed by our own EDA: ACF/PACF shows slow-decaying, long memory; the trend decomposition shows the trend range (0.031) is 16x the seasonal range (0.0019) - regime shifts dominate, exactly what RV_week/RV_month sense. All three correlate at $r = 0.58-0.61$ with the target.

ATR_X (Average True Range)

WHAT IT IS & WHAT IT MEANS

True Range = $\max(\text{High-Low}, |\text{High-PrevClose}|, |\text{Low-PrevClose}|)$ - catches gaps a simple range misses (Wilder, 1978). ATR_X = mean(True Range, 14h) / Close: scale-free, comparable across any price level.

WHY WE NEED IT

The single strongest feature in the project - $r = 0.685$ with the target, ahead of every HAR horizon. Gives the model its most immediate, hour-by-hour read on current turbulence.

Feature Engineering: BB_width & What Got Dropped

Not every engineered feature makes the cut

BB_width (Bollinger Band Width)

WHAT IT IS & WHAT IT MEANS

upper/lower = $MA_{20h} \pm 2 \times std_{20h}$ (Bollinger, 1980s). $BB_width = (upper - lower) / MA_{20h}$, as a percentage. Widens in choppy trading, squeezes in quiet trading - the classic "squeeze" before a big move.

WHY WE NEED IT

Correlates at $r = 0.557$ with the target - kept because it measures dispersion around a trend over a 20h window, not ATR_X 's bar-by-bar range. Related to ATR_X ($r = 0.82$) but not redundant - both carry independent weight in every model.

Tested and Dropped

WHAT IT IS & WHAT IT MEANS

Three more candidates, tested the same rigorous way: KC_width (Keltner Channel width - ATR-based envelope), $SKEW_X$ (realized skewness - do returns lean up or down), and RVI (Relative Volatility Index - up-move vs. down-move volatility).

WHY WE NEED IT

KC_width : $r = 0.993$ with ATR_X - near-duplicate, dropped. $SKEW_X$ ($r = -0.07$) and RVI ($r = -0.009$): essentially zero correlation - dropped. Reporting what didn't work is part of an honest feature-selection process, not just what did.

EDA: Temporal Features

Is autocorrelation present - and how was it handled?

Diagnosing Autocorrelation

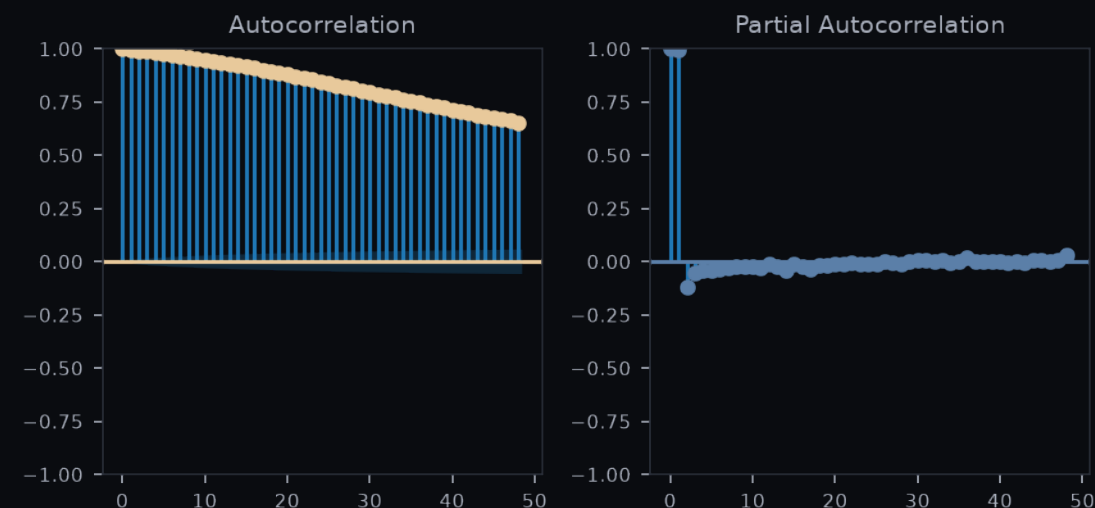
PRESENT - CONFIRMED

The ACF decays slowly across many lags instead of dropping to zero, and the ADF test rejects a unit root (stat = -12.76, $p < 0.001$) - the series is stationary but has real memory. This is genuine volatility clustering, not noise.

HOW IT WAS RESOLVED

Autocorrelated data breaks the i.i.d. assumption behind random splits and K-fold CV - a random shuffle would leak adjacent-in-time rows into training. Resolved with a chronological train/test split and TimeSeriesSplit CV (Preprocessing), plus lagged features - and by using the autocorrelation itself as HAR-RV-X's core signal, not discarding it.

ACF / PACF (48 Lags)

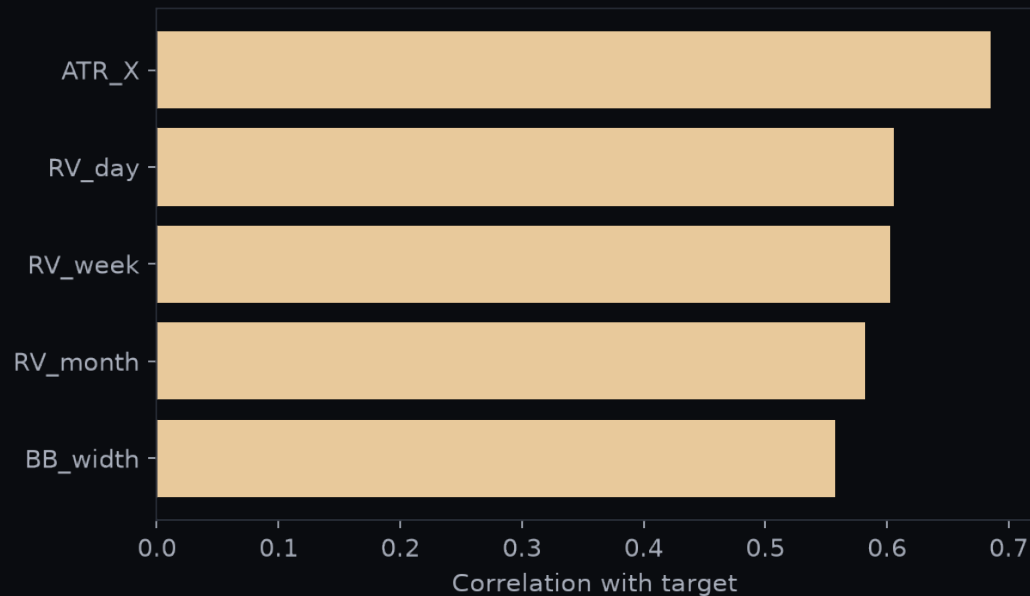


TAKEAWAY

Autocorrelation isn't a nuisance to remove here - it's the predictive signal. HAR-RV-X's lagged, multi-horizon regressors exploit it directly, while chronological splitting keeps evaluation honest.

Feature Selection: Correlation & Chi-Square

Correlation Analysis: every continuous feature kept for modeling is meaningfully correlated with the target - none are statistical accidents.



Chi-Square: Calendar Features

HOURLY OF DAY - NOT SIGNIFICANT

$\chi^2 = 3.82$, $p = 1.0$ - no relationship with the volatility regime. Not used as a feature, in any model.

DAY OF WEEK - SIGNIFICANT

$\chi^2 = 2886.3$, $p < 0.001$ - a real weekly rhythm. Mon/Sun skew toward low volatility, Tue-Fri skew toward moderate/high. Added as $\text{dow_sin}/\text{dow_cos}$ to every model.

STRONGEST FEATURE

ATR_X leads the continuous features at $r = 0.685$ with the target - the single best predictor kept in the final feature set.

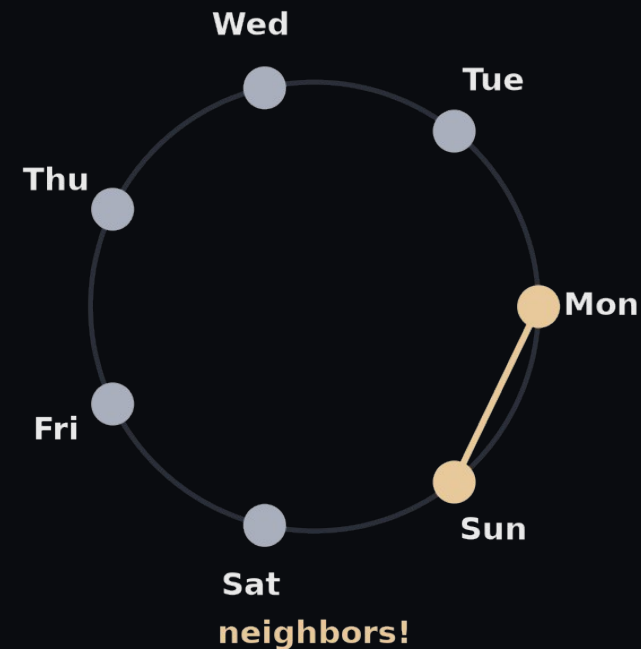
Using Time as a Feature: dow_sin & dow_cos

Making sure the model reads Sunday and Monday as neighbors

Raw day number (0-6)



dow_sin / dow_cos (on a circle)



IN XGBOOST

- `add_time_features()` builds `hour_sin`, `hour_cos`, `dow_sin`, `dow_cos` from Date

THE RESULT

- day-of-week: kept ($p < 0.001$). hour-of-day: dropped ($p = 1.0$)

Using Time as a Feature: `dow_sin` & `dow_cos`

Making sure the model reads Sunday and Monday as neighbors

- 1 The problem: raw day numbers confuse a model - Sunday (6) and Monday (0) look far apart, even though they're next to each other on the calendar
- 2 The fix: place each day on a circle using sine and cosine - `dow_sin` and `dow_cos`
- 3 Now Sunday and Monday sit right next to each other, just like on a real calendar
- 4 In XGBoost specifically: an `add_time_features()` function builds all 4 cyclical columns - `hour_sin`, `hour_cos`, `dow_sin`, `dow_cos` - from the Date column, and adds them on top of the HAR features just for XGBoost's own feature matrix
- 5 Chi-square test: day-of-week is real ($p < 0.001$) - kept. Hour-of-day is not ($p = 1.0$) - dropped

THE RESULT

`dow_sin` and `dow_cos` became real features in HAR-RV-X and XGBoost - not just tested and thrown away. `hour_sin`/`hour_cos` stayed in XGBoost too, but ended up with almost zero importance, exactly as the chi-square test predicted.

Preprocessing

Preparing a leak-free feature matrix and target for modeling

Preprocessing Steps

- 1 Build target: volatility_2d shifted 48h (HORIZON) into the future
- 2 HORIZON set equal to the volatility window - zero overlapping hours
- 3 Chronological split at a fixed date - never randomly shuffled
- 4 TimeSeriesSplit (5-fold) for every hyperparameter search - never random K-fold

Train/Test Split

TRAIN SET

55,821 rows

2017-09 to 2024-01

TEST SET

22,602 rows

2024-02 to 2026-08

WHY CHRONOLOGICAL

Time-series data is autocorrelated - a random shuffle would leak information from adjacent-in-time rows into training. Every split (train/test, and every cross-validation fold) respects chronological order.

ZERO-OVERLAP TARGET

HORIZON (48h) equals the volatility window (48h) on purpose, so the most recent known feature value and the target share zero overlapping hours - no leakage.

Baseline: GARCH

Formula and Parameter Meanings

FORMULA

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

WHAT EACH PARAMETER MEANS

- σ_t^2 = the variance being solved for (today's estimate)
- ω = baseline/floor variance level
- α = weight on yesterday's squared shock - how reactive the model is
- β = weight on yesterday's own variance estimate - how much memory it has
- ε_{t-1} = the previous shock (return surprise)
- Fitted here: $\omega=0.0138$, $\alpha=0.10$, $\beta=0.88$ ($\alpha+\beta=0.98$ - very persistent)

CHARACTERISTICS

- Textbook, most-cited baseline for volatility forecasting
- Sets the performance bar every other model here must clear
- Forecast mean-reverts within roughly 34 hours ($\alpha+\beta = 0.98$)
- By 48 hours, has already lost most of its ability to react to current conditions
- Reflected directly in its badly negative R^2 (-1.62)

Baseline: GARCH

Standard GARCH(1,1) with Student-t Errors

FORMULA

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Model Configuration

Vol Model	GARCH(1,1)
Distribution	Student's t
Fit on	Hourly returns
Forecast horizon	48 hours

Performance (Test Set)

R2 (Test Set)

-1.62

RMSE

0.003342

MAE

0.003042

KEY INSIGHT

GARCH's forecast mean-reverts toward a long-run average well before 48 hours are up - a structural weakness at this horizon, not an implementation bug. It sets the bar every required model below must clear.

Model 1: EWMA

Formula and Parameter Meanings

FORMULA

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

WHAT EACH PARAMETER MEANS

- σ_t^2 = today's variance estimate (what's being solved for)
- λ = the decay factor - the model's only tunable parameter
- σ_{t-1}^2 = yesterday's variance estimate
- r_{t-1}^2 = yesterday's squared return (the newest shock)
- Tuned value: $\lambda=0.99$ - barely reacts to any single hour, stays near a smooth average

CHARACTERISTICS

- Useful midpoint in complexity between GARCH and a full regression
- One dial, no distributional assumptions, no feature engineering
- Tuned via TimeSeriesSplit CV (best $\lambda = 0.99$)
- Comfortably clears the GARCH baseline ($R^2 = 0.140$ vs -1.618)
- No exogenous features - caps how far it can go on its own

Model 1: EWMA

RiskMetrics-Style Exponential Smoothing (Tuned)

FORMULA

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

Model Configuration

Named After	RiskMetrics (J.P. Morgan, 1996)
Lambda (Tuned)	0.99
Search Range	0.85 - 0.995
Selection	TimeSeriesSplit CV

Performance (Test Set)

R2 (Test Set)

0.140

RMSE

0.001916

MAE

0.001434

KEY INSIGHT

A single, fully transparent parameter - no hidden layers, nothing opaque. Clears the GARCH baseline by a wide margin, but its recursive, features-free design caps how much further it can improve.

Model 2: HAR-RV-X

Formula and Parameter Meanings

FORMULA

$$RV_{t+h} = \beta_0 + \beta_1 RV_{day} + \beta_2 RV_{week} + \beta_3 RV_{month} + \beta_4 ATR_X + \beta_5 BB_{width} + \beta_6 dow_{sin} + \beta_7 dow_{cos} + \varepsilon$$

WHAT EACH PARAMETER MEANS

- β_0 = intercept (baseline level)
- $\beta_1 \dots \beta_7$ = regression coefficients - how strongly each feature moves the forecast
- $RV_{day/week/month}$ = realized volatility at 3 lookback windows
- ATR_X , BB_{width} = technical indicators
- dow_{sin}/dow_{cos} = cyclical day-of-week
- ε = residual error
- Tuned via Ridge, $\alpha=0.01$ - light regularization

CHARACTERISTICS

- Natural fit for our EDA: strong autocorrelation and a large trend component
- Best-performing model of the 3 required models ($R^2 = 0.310$)
- Edges out XGBoost once day-of-week was added as a feature
- Day-of-week confirmed significant by chi-square ($\chi^2 = 2886$, $p < 0.001$)

Model 2: HAR-RV-X

Ridge-Regularized Regression on RV Horizons (Tuned)

FORMULA

$$RV_{t+h} = \beta_0 + \beta_1 RV_{day} + \beta_2 RV_{week} + \beta_3 RV_{month} + \beta_4 ATR_X + \beta_5 BB_{width} + \beta_6 dow_{sin} + \beta_7 dow_{cos} + \varepsilon$$

Model Configuration

Regressors	7 (RV horizons, ATR_X, BB_width, dow_sin/cos)
Regularization	Ridge
Alpha (GridSearchCV)	0.01
CV	TimeSeriesSplit, 5-fold

Performance (Test Set)

R2 (Test Set)

0.310

RMSE

0.001715

MAE

0.001281

KEY INSIGHT

Best of the 3 required models. Adding day-of-week (dow_sin/dow_cos), backed by a significant chi-square result, lifted R2 from 0.234 to 0.31 - the single biggest jump in the whole project.

Model 3: XGBoost

Formula and Parameter Meanings

FORMULA

$\hat{y} = \sum f_k(x)$ — an ensemble of shallow regression trees

WHAT EACH PARAMETER MEANS

- $\hat{y}$ = the predicted value
- $f_k(x)$ = the k-th tree's contribution, built to correct trees 1...k-1's errors
- x = the feature vector (HAR features + calendar)
- K ($n_estimators$) = number of trees in the ensemble
- Tuned via GridSearchCV: $n_estimators=100$, $max_depth=3$, $learning_rate=0.05$

CHARACTERISTICS

- Tests whether the feature-volatility relationship is genuinely nonlinear
- Clearest overfitting story in the project
- Default settings overfit badly ($R^2 = -0.14$)
- GridSearchCV-tuned shallow trees fix that ($R^2 = 0.306$)
- Sharpest before/after tuning contrast in the whole notebook

Model 3: XGBoost

Gradient-Boosted Trees, Tuned with GridSearchCV

FORMULA

$\hat{y} = \sum f_k(x)$ — an ensemble of shallow regression trees

Model Configuration

Regressors	HAR features + cyclical hour/day-of-week
n_estimators	100
max_depth / learning_rate	3 / 0.05
CV	TimeSeriesSplit, 5-fold

Performance (Test Set)

R2 (Test Set)

0.306

RMSE

0.001721

MAE

0.001279

KEY INSIGHT

Untuned XGBoost overfits badly (R2 = -0.14) - GridSearchCV's shallow, slow-learning trees fix that. Slightly behind HAR-RV-X here once day-of-week was added to both models.

Model Comparison

Every Model Tried in the Final Notebook, Test-Set Performance

Model	RMSE	R2	MAE	Type
GARCH (baseline)	0.003342	-1.618	0.003042	Baseline
EWMA (base, lambda=0.94)	0.002191	-0.125	0.001620	Model 1 - base
EWMA (tuned, lambda=0.99)	0.001916	0.140	0.001434	Model 1 - tuned
HAR-RV-X (base, LinearReg)	0.001714	0.311	0.001278	Model 2 - base
HAR-RV-X (tuned, Ridge)	0.001715	0.310	0.001281	Model 2 - tuned
XGBoost (base, default)	0.002204	-0.139	0.001493	Model 3 - base
XGBoost (tuned, GridSearchCV)	0.001721	0.306	0.001279	Model 3 - tuned
HAR-RV-X (log-target)	0.001732	0.297	0.001212	Extra attempt
XGBoost (log-target)	0.001698	0.324	0.001172	Extra - best overall
Ensemble (HAR-RV-X + XGBoost)	0.001710	0.315	0.001274	Extra attempt
XGBoost (PowerTransformer)	0.001705	0.319	0.001173	Extra attempt
EWMA (log-space tuned)	0.001916	0.140	0.001434	Extra attempt
Facebook Prophet (base)	0.002604	-0.589	0.001905	Extra attempt
Facebook Prophet (tuned)	0.003429	-1.755	0.002780	Extra - worst overall
HAR-RV-X (detrended via RV_month)	0.001713	0.313	0.001274	Extra attempt

KEY FINDING

XGBoost (log-target) is the best number overall (R2=0.324); HAR-RV-X (tuned) is the best of the 3 required models (R2=0.310); Facebook Prophet (tuned) is the single worst result.

Limitations

Category 01

Data Constraints

- No fundamental or on-chain signals - only price/volume history.
- News, regulation, and macro shocks are invisible to the model.
- Single asset - no cross-crypto spillover signal used.

Category 02

Modeling Constraints

- GARCH's mean-reversion is structural, not tunable away.
- Chi-square confirmed day-of-week matters but hour-of-day doesn't.
- Time constraints limited exhaustive deep-learning exploration.

Category 03

Statistical Ceiling

- Even the strongest feature (ATR_X) has only $r = 0.685$ with the target.
- Models cannot exceed the information content of their inputs.
- A moderate R^2 ceiling is a predictable result, not a bug.

KEY TAKEAWAY

Meaningful improvement needs genuinely new information (news sentiment, on-chain metrics), not further tuning of what's already here. Neither more model complexity nor more compute alone is sufficient.

Operational Implications

Where a 0.31 R2 forecast is genuinely useful - and where it is not.

Appropriate Uses

Position Sizing

Scale trade size to the forecast - bigger expected move, smaller position.

Options Pricing Input

One data-driven volatility estimate to compare against market pricing.

Risk Reporting

A continuous VaR input for short-horizon risk reviews, not a single trade call.

Inappropriate Uses

Sole Trading Signal

Explains ~31% of variance - never trade on this number alone.

Real-Time Execution

Not built or tested for split-second, automated trade triggers.

Certainty in Legal/Compliance Filings

A forecast with real error bars, not a guarantee.

One input, not the final answer.

Thank You

Any Questions?